Equivalent Fractions

Cut every piece the same way: more pieces, same amount.

2/3: cut every piece into 4



$$\frac{2}{3} = \frac{8}{12}$$

4 times as many pieces, and 4 times as many shaded.

Multiply the top and the bottom by the same number

top: $2 \times 4 = 8$

bottom: $3 \times 4 = 12$

$$\frac{2}{3} = \frac{8}{12}$$

Divide to get bigger pieces



$$\frac{6}{10} = \frac{3}{5}$$

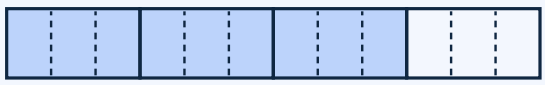
Put the pieces in pairs: top ÷ 2 and bottom ÷ 2.

Hundredths: tenths cut into 10

$$\frac{3}{10} = \frac{30}{100}$$

Every tenth is 10 hundredths.

Your turn!



$$\frac{3}{4} = \frac{9}{12}$$

For teachers

Goal: equivalent fractions by multiplying or dividing the top and bottom by the same number (4.NF.1). Say "Cut every piece into 4. How many pieces now? How many are shaded?"

Equivalent Fractions: Practice

Do the same to the top and the bottom.

Part 1 Cut the pieces Count the small pieces, then the shaded ones.

1) Each piece is cut into 4.



$\frac{1}{3} = \frac{4}{12}$

2) Each piece is cut into 2.



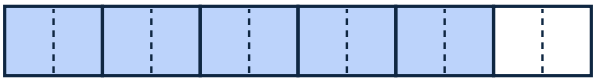
$\frac{3}{5} = \frac{6}{10}$

3) Each piece is cut into 3.



$\frac{2}{4} = \frac{6}{12}$

4) Each piece is cut into 2.



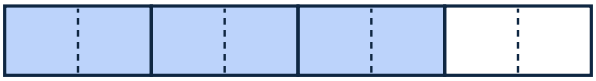
$\frac{5}{6} = \frac{10}{12}$

5) Each piece is cut into 6.



$\frac{1}{2} = \frac{6}{12}$

6) Each piece is cut into 2.



$\frac{3}{4} = \frac{6}{8}$

Part 2 Times the top and the bottom Multiply both by the number shown.

7) times 2: $\frac{2}{5} = \frac{4}{10}$

8) times 3: $\frac{1}{4} = \frac{3}{12}$

Equivalent Fractions: Practice (continued)

Part 2 Times the top and the bottom (continued)

9) times 2: $\frac{5}{6} = \frac{10}{12}$

10) times 4: $\frac{2}{3} = \frac{8}{12}$

11) times 4: $\frac{1}{3} = \frac{4}{12}$

12) times 10: $\frac{3}{10} = \frac{30}{100}$

13) times 6: $\frac{1}{2} = \frac{6}{12}$

14) times 20: $\frac{4}{5} = \frac{80}{100}$

15) times 2: $\frac{1}{6} = \frac{2}{12}$

16) times 10: $\frac{9}{10} = \frac{90}{100}$

Part 3 Find the top number No help this time.

17) $\frac{3}{4} = \frac{9}{12}$

18) $\frac{1}{3} = \frac{4}{12}$

19) $\frac{5}{6} = \frac{10}{12}$

Equivalent Fractions: Practice (continued)

Part 3 Find the top number (continued)

20) $\frac{1}{4} = \frac{\boxed{3}}{12}$

21) $\frac{7}{10} = \frac{\boxed{70}}{100}$

22) $\frac{1}{2} = \frac{\boxed{50}}{100}$

23) $\frac{2}{3} = \frac{\boxed{8}}{12}$

24) $\frac{3}{5} = \frac{\boxed{60}}{100}$

25) $\frac{1}{6} = \frac{\boxed{2}}{12}$

26) $\frac{1}{4} = \frac{\boxed{25}}{100}$

27) $\frac{1}{2} = \frac{\boxed{6}}{12}$

28) $\frac{2}{5} = \frac{\boxed{40}}{100}$

Part 4 Find the bottom number No help this time.

29) $\frac{3}{4} = \frac{75}{\boxed{100}}$

30) $\frac{2}{3} = \frac{8}{\boxed{12}}$

31) $\frac{1}{5} = \frac{20}{\boxed{100}}$

Equivalent Fractions: Practice (continued)

Part 4 Find the bottom number (continued)

32) $\frac{5}{6} = \frac{10}{12}$

33) $\frac{3}{10} = \frac{30}{100}$

34) $\frac{1}{2} = \frac{6}{12}$

35) $\frac{3}{4} = \frac{9}{12}$

36) $\frac{4}{5} = \frac{80}{100}$

37) $\frac{9}{10} = \frac{90}{100}$

Part 5 Fewer, bigger pieces No help this time. Divide the top and the bottom by the same number.

38) $\frac{8}{12} = \frac{2}{3}$

39) $\frac{60}{100} = \frac{3}{5}$

40) $\frac{9}{12} = \frac{3}{4}$

41) $\frac{50}{100} = \frac{1}{2}$

42) $\frac{10}{12} = \frac{5}{6}$

43) $\frac{40}{100} = \frac{2}{5}$

Equivalent Fractions: Practice (continued)

Part 5 Fewer, bigger pieces (continued)

44) $\frac{4}{12} = \frac{\boxed{1}}{3}$

45) $\frac{75}{100} = \frac{\boxed{3}}{4}$

46) $\frac{6}{12} = \frac{\boxed{1}}{2}$

47) $\frac{20}{100} = \frac{\boxed{1}}{5}$

48) $\frac{2}{12} = \frac{\boxed{1}}{6}$

49) $\frac{70}{100} = \frac{\boxed{7}}{10}$

Part 6 Equal or not? Circle yes or no.

50) $\frac{2}{3}$ and $\frac{8}{12}$ — (yes) / no

51) $\frac{3}{4}$ and $\frac{6}{12}$ — yes / (no)

52) $\frac{1}{5}$ and $\frac{20}{100}$ — (yes) / no

53) $\frac{5}{6}$ and $\frac{10}{12}$ — (yes) / no

54) $\frac{2}{5}$ and $\frac{25}{100}$ — yes / (no)

55) $\frac{3}{10}$ and $\frac{30}{100}$ — (yes) / no

Equivalent Fractions: Practice (continued)

Part 6 Equal or not? (continued)

56) $\frac{1}{4}$ and $\frac{4}{12}$ — yes / **no**

57) $\frac{7}{10}$ and $\frac{70}{100}$ — **yes** / no

Part 7 Equal fractions

58) $\frac{3}{5} = \frac{9}{15}$

59) $\frac{1}{4} = \frac{4}{16}$

60) $\frac{2}{4} = \frac{10}{20}$

61) $\frac{3}{4} = \frac{12}{16}$

62) $\frac{4}{5} = \frac{12}{15}$

63) $\frac{3}{5} = \frac{15}{25}$

64) $\frac{2}{5} = \frac{10}{25}$

65) $\frac{1}{6} = \frac{5}{30}$

66) $\frac{3}{6} = \frac{15}{30}$

Equivalent Fractions: Practice (continued)

Part 7 Equal fractions (continued)

67) $\frac{5}{6} = \frac{\boxed{20}}{24}$

68) $\frac{1}{5} = \frac{\boxed{3}}{15}$

69) $\frac{2}{6} = \frac{\boxed{6}}{18}$

70) $\frac{1}{6} = \frac{\boxed{3}}{18}$

71) $\frac{4}{5} = \frac{\boxed{16}}{20}$

72) $\frac{1}{3} = \frac{\boxed{5}}{15}$

73) $\frac{2}{5} = \frac{\boxed{6}}{15}$

74) $\frac{1}{5} = \frac{\boxed{5}}{25}$

75) $\frac{2}{4} = \frac{\boxed{8}}{16}$

76) $\frac{4}{6} = \frac{\boxed{16}}{24}$

77) $\frac{2}{6} = \frac{\boxed{8}}{24}$

78) $\frac{1}{6} = \frac{\boxed{4}}{24}$

Equivalent Fractions: Practice (continued)

Part 7 Equal fractions (continued)

79) $\frac{2}{6} = \frac{\boxed{10}}{30}$

80) $\frac{3}{5} = \frac{\boxed{12}}{20}$

81) $\frac{3}{6} = \frac{\boxed{9}}{18}$

Part 8 Make it simpler

82) $\frac{20}{24} = \frac{\boxed{5}}{6}$

83) $\frac{12}{16} = \frac{\boxed{3}}{4}$

84) $\frac{4}{24} = \frac{\boxed{1}}{6}$

85) $\frac{12}{20} = \frac{\boxed{3}}{5}$

86) $\frac{3}{18} = \frac{\boxed{1}}{6}$

87) $\frac{9}{15} = \frac{\boxed{3}}{5}$

88) $\frac{15}{18} = \frac{\boxed{5}}{6}$

89) $\frac{3}{15} = \frac{\boxed{1}}{5}$

90) $\frac{4}{20} = \frac{\boxed{1}}{5}$

Equivalent Fractions: Practice (continued)

Part 8 Make it simpler (continued)

91) $\frac{8}{20} = \frac{2}{5}$

92) $\frac{12}{15} = \frac{4}{5}$

93) $\frac{4}{16} = \frac{1}{4}$

94) $\frac{6}{15} = \frac{2}{5}$

95) $\frac{16}{20} = \frac{4}{5}$

Part 9 Remember? Problems from earlier packets.

96) $1{,}248 \div 4 = 312$

97) $42 \times 13 = 546$

98) $784 \div 4 = 196$

99) $26 \times 3 = 78$

100) 5,912 to the nearest hundred = 5,900

101) $\frac{4}{5}$ of $25 = 20$

Equivalent Fractions: Practice (continued)

Part 10 Word problems Draw a strip and cut the pieces.

102) Mia ate 3/4 of a pizza cut in 12 slices. How many slices did she eat?

Number sentence 3/4 = 9/12

Answer 9 slices

Draw or show your work

103) A fence has 100 boards. 3/10 of them are painted. How many boards are painted?

Number sentence 3/10 = 30/100

Answer 30 boards

Draw or show your work

104) Leo read 8/12 of his book. Kim read 2/3 of the same book. Did they read the same amount?

Number sentence 8/12 = 2/3

Answer yes

Draw or show your work

105) 60 of 100 seats are full. What fraction is that, in fifths?

Number sentence 60/100 = 3/5

Answer 3/5 full

Draw or show your work

106) Challenge: Sam says 2/5 = 4/10 = 40/100. Is he right? What would 2/5 be in twentieths?

Number sentence 2/5 = 8/20

Answer 8/20

Draw or show your work

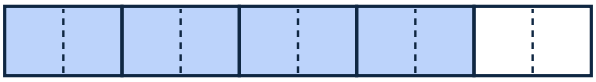
Check-In: Equivalent Fractions

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Cut the pieces

1) Each piece is cut into 2.



$\frac{4}{5} = \frac{8}{10}$

Part 2 Find the top number

2) $\frac{4}{6} = \frac{8}{12}$

3) $\frac{3}{6} = \frac{6}{12}$

4) $\frac{4}{5} = \frac{80}{100}$

Part 3 Fewer, bigger pieces

5) $\frac{50}{100} = \frac{2}{4}$

6) $\frac{60}{100} = \frac{6}{10}$

7) $\frac{6}{12} = \frac{2}{4}$

Part 4 Equal or not? Circle yes or no.

8) $\frac{4}{6}$ and $\frac{8}{12}$ (yes) no

9) $\frac{2}{4}$ and $\frac{5}{8}$ yes (no)